



Q1. Is there any health risk with a Wi-Fi router and what would be a safe distance from a Wi-Fi router installed in a building?

R. Roy

A1. There are many articles on the Internet suggesting that a Wi-Fi router poses potential health risks while others negate it. Scientists and researchers are still studying the effects of long-term electromagnetic radiation.

Some experts say that wireless routers and allied systems produce radiation in low GHz frequency, which is potentially dangerous even when it is within the so-called 'safe limits' set by some countries. As the RF radiations could still cause harmful or non-thermal biological problems, we should understand the risks associated with Wi-Fi radiation from our own routers and those installed by our neighbours, offices, or even in public places.

Wi-Fi routers generally use 2.4GHz or 5GHz band. The 5GHz Wi-Fi routers offer higher download speeds but the 2.4GHz waves can travel through walls, which 5GHz have difficulty in doing.

The radiation from a router is the strongest near its antenna and reduces when you move away. Most experts suggest that radiation level of three to six microwatts per metre squared are safe, which can be measured using an RF/EMF meter.

You can use a meter to measure the Wi-Fi radiation in your environment or you can use Wi-Fi router guards available in the market if you are feeling stressed due to the radiation.

Q2. What is fanout in PCB layout design? Can we place a via under the BGA pad without fanout?

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A2. In PCB layout designing, especially for BGA (ball grid array), the PCB fanout, land, and via are particularly important. Fanouts are traces from the device lands to adjacent via, as shown in Fig. 1. A via is an electrical connection between layers in PCB to connect the input and output, power, and ground tracks.

Typically, there is one via per land. PCB lands are where device balls sit and get soldered to. One of the important and difficult aspects of PCB design using fine pitch BGAs is the layout of their lands and fanouts.

You can place a via under the BGA pad without fanout, but it comes with advantages and disadvantages. Having vias placed on a pad is a common practice. Some designers place

vias next to a pad, in some cases even by removing some pads from the BGA footprint. Here fanouts are not involved.

If there is a via in pad (Fig. 2), it needs to be filled either with copper or non-conductive material. Because an open hole would cause the solder to flow away from the pad, which is not desirable. This via creates an extra step in PCB manufacturing and involves additional costs. That is why many designers don't use this option.

The second problem is the required drill-hole diameter. For example, it is difficult to place a 0.2mm hole in a pad of a 0.4mm spaced BGA. Using a smaller drill size would result in higher costs. One of the solutions is to use normal fanout routing with few vias. This option would give you a much cheaper solution.

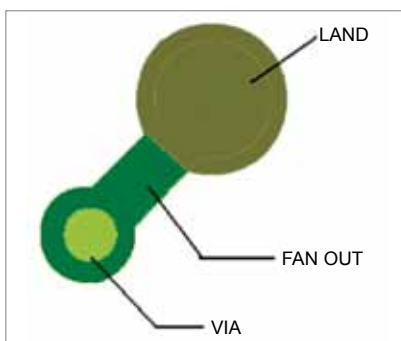


Fig. 1: Typical fanout in BGA PCB
(Credit: www.nexlogic.com)



Fig. 2: Via in Pad (Credit: www.nexlogic.com)